Presentation of PCI & the PCI Sustainability project



- Introductory words from the Belmont Forum
- Introduction of PCI by Thomas Guillemaud (INRAE) 20 min
- Introduction of PCI Sustainability by Eric Pante (CNRS) 10 min
- Questions / Discussion 30 min

Eric PANTE, CNRS | Oct 15, 2026

*something different is surely "in the air," something that is intellectually exciting, practically compelling, and might as well be called "**sustainability science**."*

Clark & Dickson 2003 PNAS

Good morning / evening / afternoon everyone, and thank you for attending the seminar which aims to present both the peer community in system and our project to create a new thematic PCI called PCI sustainability. First, we will hear a few words from Hilario Espinosa from the Belmont Forum Secretariate, then have an introduction of the PCI system by one of its co-creators, Thomas Guillemaud the creators of PCI and then I will introduce the PCI sustainability project please keep your questions until the end of the talk, as we will have plenty of time to discuss. *Finally, I will record the presentation to make it available online, but the only the presentation screen and the presentation themselves will be shared.* Thank you again, and the mic is yours Hilario !

Why a new thematic PCI ?



1- Because science for sustainability is not sustainability science

Sustainability science is […] a vibrant arena that is bringing together scholarship and practice, global and local perspectives from north and south, and disciplines across the natural and social sciences, engineering, and medicine.
Clark & Dickson 2003 PNAS

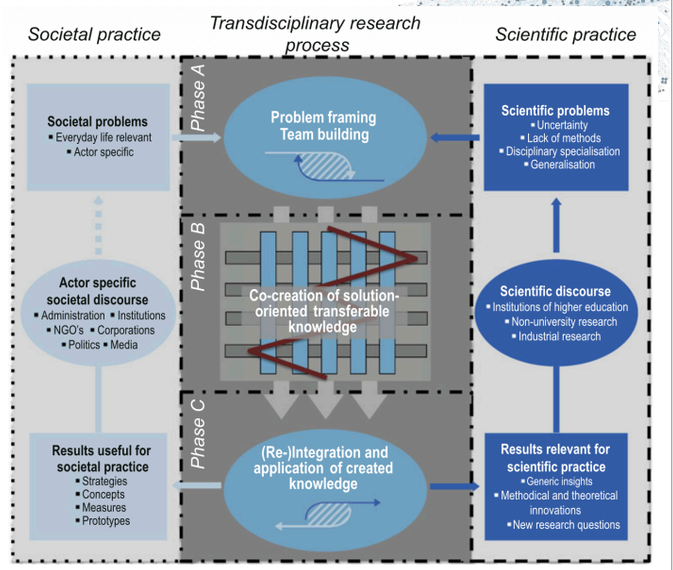
> *Sustainability science seeks to extend scientific investigation into domains characterized by a distinct problem-solving agenda, physical and social complexity, and complex moral and ethical landscapes. In this endeavor, it arguably pushes scientific investigation beyond its usual comfort zones, raising fundamental issues about how best to structure such investigation.*
> Nagatsu et al. 2020

I will now introduce the project of creating a new thematic PCI, called PCI sustainability. As you all must probably know science for sustainability is not sustainability science. while science for sustainability is disciplinary, and fits multiple of the existing thematic PCI, sustainability science is relatively new discipline, about 20 years old, that is problem oriented, and seeks perspectives from diverse disciplines and context. As Nagasu et al put it, *sustainability science, arguably pushes scientific investigation beyond its usual comfort zones*.

Why a new thematic PCI ?

2- Because sustainability science has its own epistemological framework

sustainability challenges require new ways of knowledge production and decision-making. One key aspect of sustainability science, therefore, is the involvement of actors from outside academia into the research process in order to integrate the best available knowledge, reconcile values and preferences, as well as create ownership for problems and solution options.
Lang et al 2012

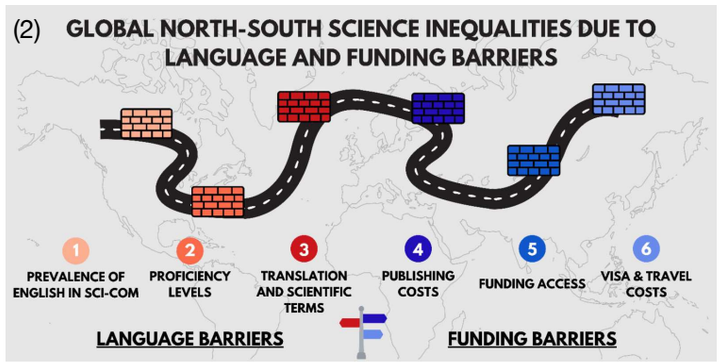


As sustainability science developed, so did it own epistemological framework as exemplified here for the transdisciplinary research process, which is core to sustainability science. This figure shows how societal problems, frame scientific problems, and the feedback loop that is created between scientific practice and societal practice. Within this framework, sustainability scientists integrate the best available knowledge, reconcile values and preferences, and create ownership for problems and solution options.

Why a new thematic PCI ?



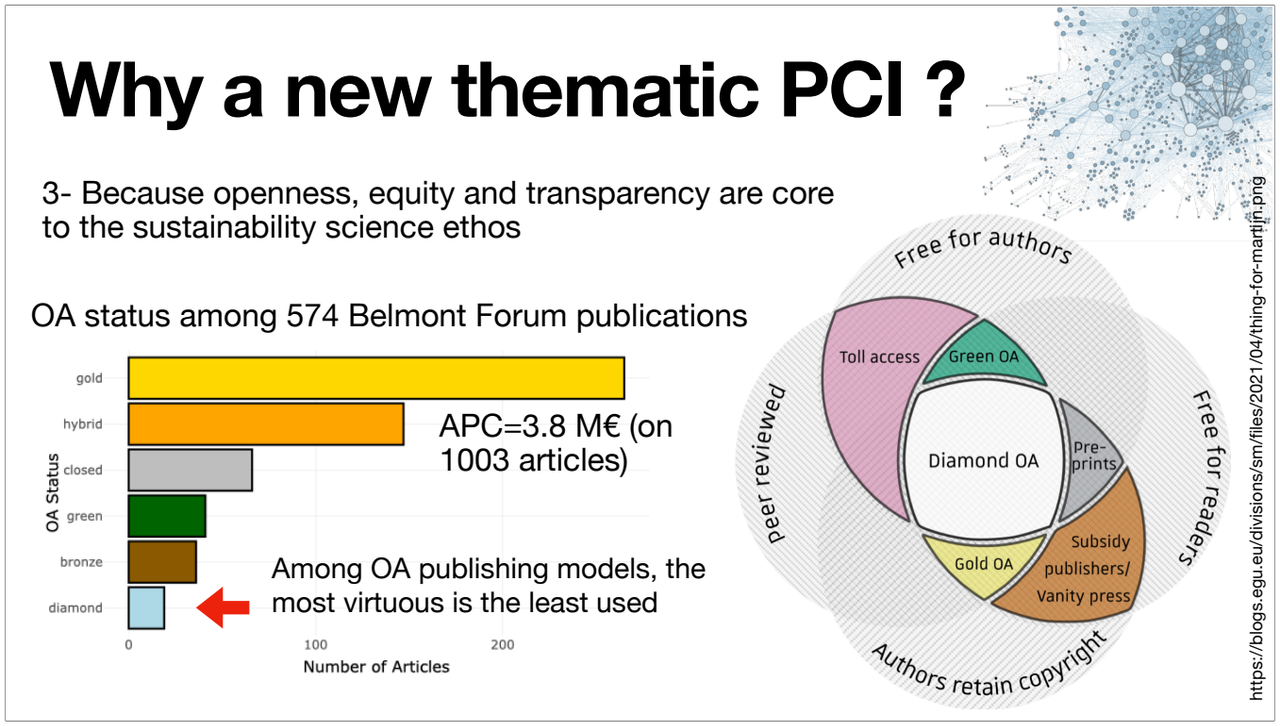
3- Because openness, equity and transparency are core to the sustainability science ethos



- Yet, it is a challenge to publish OA in Sustainability science, particularly in the Global South
- “78% of research cited in SDG policy includes authors based only in countries in the Global North” (1)
- "Global South policy remains highly dependent on Global North research” (1)
- “Publication models can influence SDG policy impact:” policymakers from the Global South cite OA research more frequently than non-OA. (1)

(1) From publications to policy: The impact of research towards achieving the Sustainable Development Goals. Springer Nature, Overton
(2) Turba R et al (2026) Global North-South science inequalities due to language and funding barriers. Peer Community Journal, Vol6, e9

Because / or thanks to this transdisciplinarity, and therefore the weaving of different perspectives, narratives, knowledge production systems, sustainability science is based on core values, such as openness, equity, and transparency.
Yet, paradoxically it is a challenge to publish open access sustainability science, particularly for colleagues in the global South. And this was recently studied and published in a report, signed by Springer nature and Overton, who found that 78% of the research site in SDG policies include authors based in global north countries while global self policy remains highly dependent on global North research. Springer nature and Overton also found that publication models significantly influence SDG policy impact policy makers from the global South cite open access research more frequently than non-open access research so the way we choose to publish our science matters for policy.



Here are some data on this paradox from publications stemming from the Belmont Forum collaborative research actions (which are transnational funding calls for sustainability research). Within a corpus of 574 publications, about 90% were found to be behind some kind of paywall, while only a handful were published in diamond open access which is the most virtuous publishing system as illustrated on the figure here. Diamond open access to peer-reviewed papers is free for authors, free for readers, while allowing authors to retain their copyright.



Why a new thematic PCI ?

3- Because openness, equity and transparency are core to the sustainability science ethos

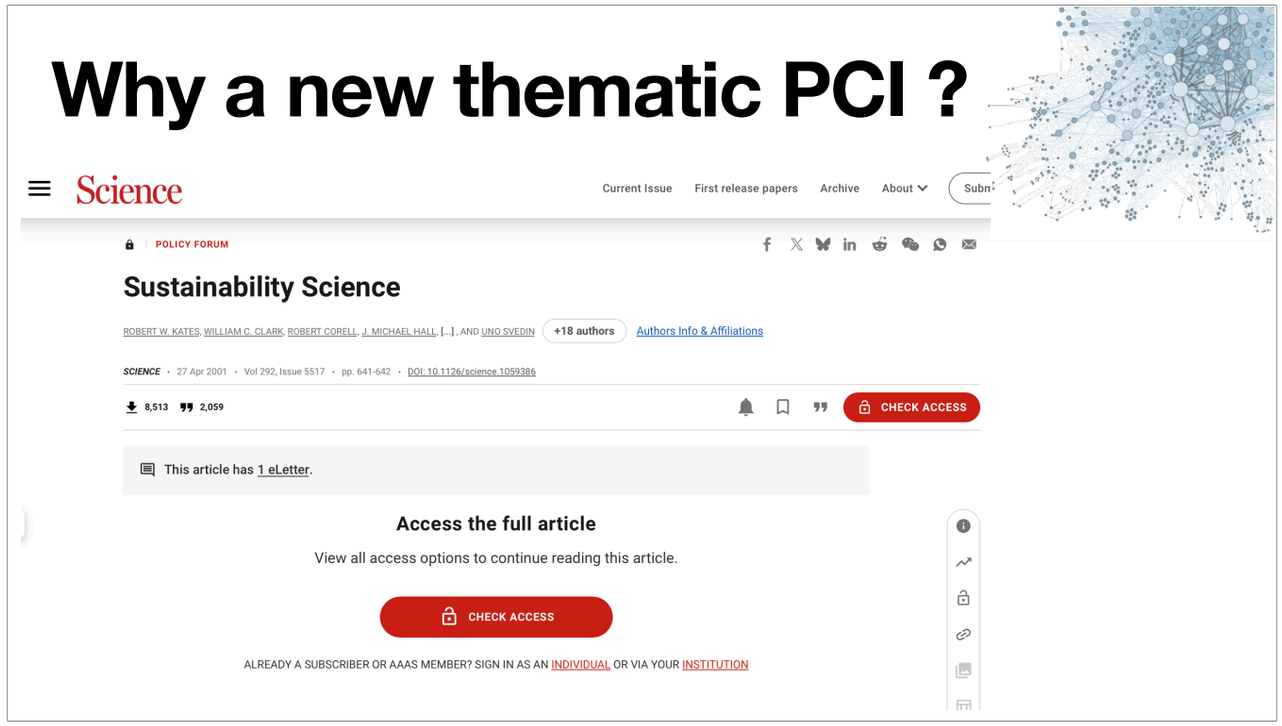
SocSES
SOCIETY FOR SOCIAL-ECOLOGICAL SYSTEMS

TABLE OF JOURNALS

Name	Type of SES Research Published	Website
Ecology & Society	Publishes interdisciplinary SES research on resilience, governance, adaptation, and transformations, with a strong emphasis on participatory approaches and knowledge co-production.	View Website
People & Nature	Explores the interactions between people and nature, integrating diverse disciplinary perspectives on human-nature relationships and sustainability.	View Website
Ecosystems and People	Focuses on ecosystem services, social-ecological resilience, and sustainability, emphasizing actionable and policy-relevant research.	View Website

List of 28 journals publishing SES research -> one is diamond OA.

The society for social ecological system put together a table of 28 journals publishing social ecological system research. You can find stable on their website. It's available to everyone. Within this list, there is only one publication that is diamond open access and it’s a niche journal for austral research. So the pattern that we observed that little research is published in diamond open access journals is not just related to scientists choices, but also linked to the science publishing ecosystem. Although I don’t show it here, it is also linked to the way scientists are evaluated and funded.



As I was preparing the stock, I actually found that one of the key foundational papers for sustainability science, the one by Robert Kates et al from 2001, in science, is actually behind a paywall.

Why a new thematic PCI ?



So what would PCI Sustainability look like?

- Encompass the inherently integrative nature of transdisciplinary sustainability research, which weaves together knowledge from natural and social scientists, policy practitioners, indigenous peoples, and local communities.
- Support the FAIR and CARE principles and transparent review of preprints, create a community of engaged reviewers across disciplinary and societal actor boundaries, thus elevating the visibility of sustainability-focused outputs.
- Offer an avenue for diamond open-access publication with Peer Community Journal

So based on these observations, I would argue that the creation of a new thematic PCI for sustainability science is justified and even urgent. This PCI would …

Why a new thematic PCI ?



Who supports the PCI Sustainability project?

BELMONT FORUM

cnrs ÉCOLOGIE & ENVIRONNEMENT

IAI

futureearth French Hub

IRD Institut de Recherche pour le Développement FRANCE

This project so far, is supported by the Belmont Forum, the ecology and environment Institute of the CNRS, the Inter-American Institute for global change research, the French hub of future Earth, and the French Institute for research for development, IRD. We are actively seeking support for from additional sustainability science institutions.

Join the PCI Sustainability community!



- ✓ met the 20 manuscript pledge target within days
- ✓ met the 50 Recommender pledge target within a month
- ✓ Signees from 32 countries
- ✓ Diverse: recommenders w/ ORCID match 4/4 domains, 14/26 fields, 30/252 subfields in OpenAlex

Global and Planetary Change	16
Nature and Landscape Conservation	5
Sociology and Political Science	5
Ecology	4
Management, Monitoring, Policy and Law	4
Plant Science	4
Environmental Engineering	3
Soil Science	3
Artificial Intelligence	2
Electrical and Electronic Engineering	2
Environmental Chemistry	2
General Agricultural and Biological Sciences	2
Genetics	2
Information Systems and Management	2
Oceanography	2
Social Psychology	2
Building and Construction	1
Civil and Structural Engineering	1
Developmental and Educational Psychology	1
Ecology, Evolution, Behavior and Systematics	1
Finance	1
General Health Professions	1
History and Philosophy of Science	1
Immunology	1
Information Systems	1
Molecular Biology	1
Nuclear and High Energy Physics	1
Pharmacy	1
Public Administration	1
Waste Management and Disposal	1

OpenAlex

So we are off to a great start, as we had 20 pledges for manuscript submissions within a few days of starting to advertise for this PCI sustainability project, and we got 50 pledges from people willing to act as recommenders within a month.
These signees are based in 32 different countries and have a wide range of expertise: for people who communicated be their ORCID, I gathered the fields of research in OpenAlex and as you can see here most are in global and planetary change, nature and landscape, conservation, sociology, and political science, ecology, management, law, but we also have people from finance, from public administration, from philosophy of science, etc.

Join the PCI Sustainability community!





Map of SES institutes



Now : need to reach out to the global Sustainability Research community

What the community (you!) can do:

- register your interest as recommender and / or author, or just ask to be kept in the loop
- Spread the information with colleagues
- Introduce us to contacts at Sustainability Research institutes

But we are not done yet: we need to continue communicating about this project, to make it a success and to mobilize as much of the community as possible, because PCI is about community re-appropriation of the scientific publication system. This is what you can do to help with the project …

Detailed steps for the creation of a PCI



✓ Preliminary discussions with the PCI team
✓ Choose one or two colleagues to set up and manage the PCI
✓ Define the scientific field covered by your PCI
✓ Establish a first group of recommenders
✓ Get submission pledges

- Submit your proposal to the PCI organisation for discussion and approval
- Recruit more recommenders
- Nominate a representative and set up a managing board
- Open the PCI publicly: receive submissions and manage evaluations
- Communicate on the PCI
- Find a managing editor

These are the detailed steps for the creation of a new PCI and this is what we've done so far. The next step is to submit our proposal to the PCI organization for discussion and approval, and then we will move on to recruiting more recommenders nominate a representative and set up a managing board open the PCI publicly communicate on the PCI and finding a managing editor.

Join the PCI Sustainability community!



Thank you ! 🙂

Registration form

PCI Sustainability

https://ericpante.github.io/projects/PCIsustainability.html

And with that want to thank you warmly for your attention and for your interest in PCI sustainability !
you can join our project or stay informed through the registration form here or following the link below, which I put in the chat. And with that I will open the floor for questions. Thank you!